

UNIVERSITY OF AGRICULTURAL SCIENCES, BANGLORE-560065

COLLEGE OF AGRICULTURE, GKVK

EAI-421: AI & ML IN AGRICULTURE (0+10)

PROJECT TITLE – MACHINE LEARNING BASED
GROUNDWATER QUALITY ANALYSIS

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INTRODUCTION

Groundwater is a critical resource for drinking, irrigation, and industrial applications, but its quality is increasingly impacted by natural processes and human activities.

Timely and accurate assessment of groundwater quality is essential for sustainable management.

This capstone project presents the development of an Artificial Intelligence (AI) and Machine Learning (ML)-based system to analyze and predict groundwater quality using real-world datasets, enhancing the efficiency of traditional evaluation methods.

PROBLEM STATEMENT

- Groundwater quality assessment is crucial for safe drinking water and agriculture.
- Traditional methods are time-consuming, costly, and lack real-time insights.
- 2018 data shows variations in pH, EC, TDS, hardness, indicating health and productivity risks.
- Need for efficient, accurate, and scalable AI-driven system to predict water quality.

Project Goal

- Develop ML model to classify groundwater as safe/unsafe, improving prediction accuracy and decision-making.

MOTIVATION

- Importance
 - Ensures safe drinking water (health risks: fluoride, arsenic, nitrate)
 - Protects agriculture productivity & soil health
 - Addresses India's water scarcity & stress
- Practical Applications
 - Precision irrigation & crop selection
 - Early contamination alerts for farmers
 - Informed policy & remediation efforts
- Relevance
 - Supports Clean Water
 - water management, agri-tech, risk assessment

OBJECTIVES

- 01.** To analyze the 2018 post-monsoon groundwater quality dataset and understand variations in key physicochemical parameters such as pH, EC, TDS, and hardness.
- 02.** To develop and apply machine learning models (Random Forest, Decision Tree, etc.) for predicting and classifying groundwater quality.
- 03.** To evaluate and compare model performance to identify the most accurate approach for groundwater quality prediction.

DATASET OVERVIEW

- Dataset source: Ground water quality 2018 post monsoon data from kaggle
- Number of records and features: 300 samples, 9 features (pH, EC, TDS, Hardness, Alkalinity, Chloride, Sulphate, Nitrate, Fluoride)
- Types of features: Numerical (physicochemical parameters)
- Preprocessing steps: Data loading, feature selection, normalization, train-test split

Feature	Type	Source
pH	Numerical (Continuous)	Field measurement
Electrical Conductivity	Numerical (Continuous)	Field measurement
Total dissolved solids	Numerical (Continuous)	Field measurement
Total hardness	Numerical (Continuous)	Field measurement
Alkalinity	Numerical (Continuous)	Field measurement
Chloride	Numerical (Continuous)	Laboratory analysis
Sulphate	Numerical(Continuous)	Laboratory analysis
Nitrate	Numerical (Continuous)	Laboratory analysis
Fluoride	Numerical (Continuous)	Laboratory analysis
Groundwater Quality Index	Numerical (Target)	Computed/Derived

APPROACH

- **Data Preprocessing**

Upload CSV/Excel dataset in Google Colab

Feature selection (pH, EC, TDS, etc.)

Normalize data using MinMaxScaler

Train-test split (80:20)

- **Exploratory Data Analysis**

Water quality distribution

Correlation heatmap

Identify key influencing parameters

- **Model Selection**

Random Forest

Decision Tree

Logistic Regression

- **Model Training & Validation**

Train models on training data

Test on unseen data

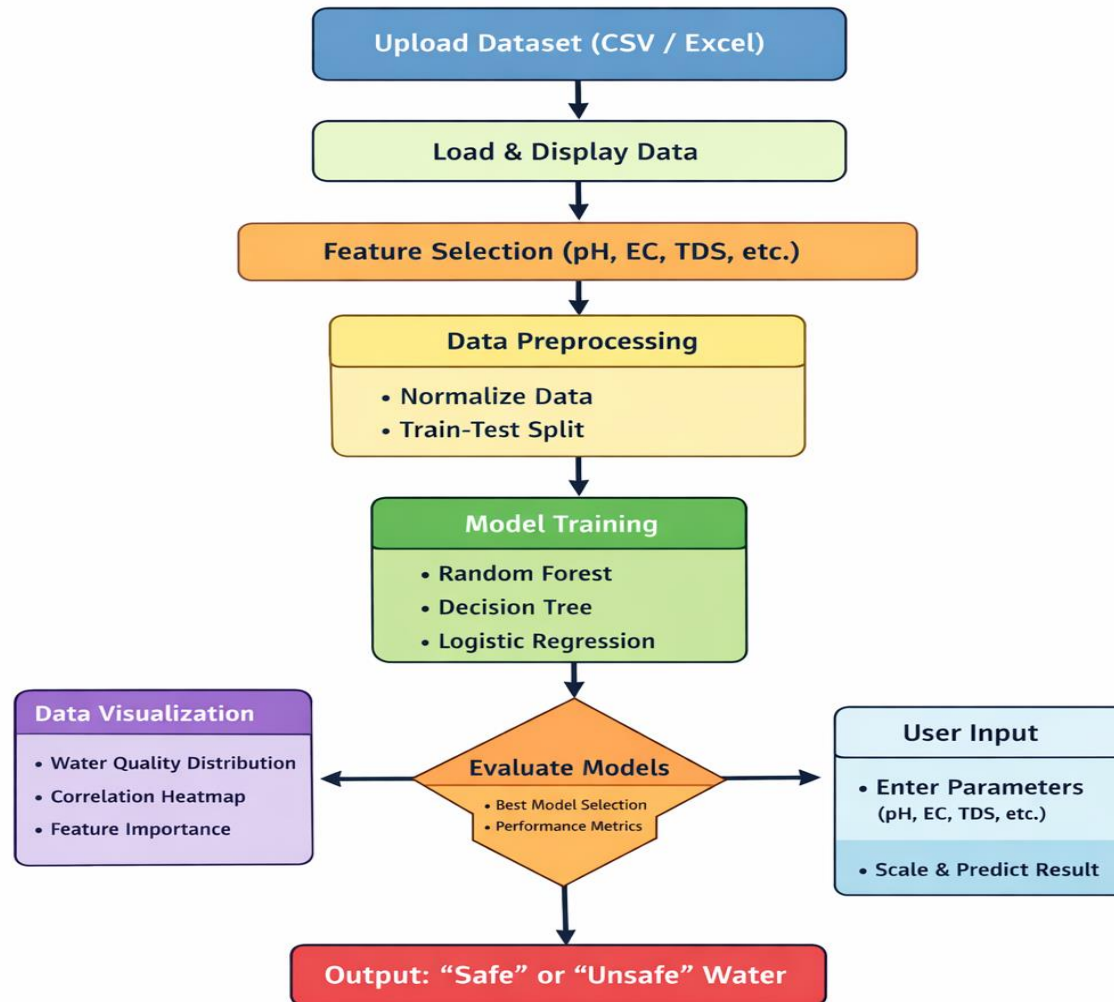
Select best-performing model

- **Evaluation Metrics**

Accuracy

F1-score

Confusion Matrix



Tool / Technology	Role in the Project
Python 3.x	Core programming language used to implement the entire ML pipeline
pandas	Used for data loading, manipulation, and preprocessing of dataset
NumPy	Performs numerical computations and array operations
matplotlib	Used for data visualization like bar charts and heatmaps
scikit-learn	Machine learning library used for model building and evaluation
Random Forest Classifier	Ensemble algorithm that improves accuracy using multiple decision trees
Decision Tree Classifier	Tree-based model that classifies data using decision rules
Logistic Regression	Statistical model used for binary classification (Safe/Unsafe)
train_test_split	Splits dataset into training and testing sets
Confusion Matrix	Evaluates classification performance by comparing actual vs predicted values

PYTHON CODE

```
# --- 1. Upload File in Colab ---  
from google.colab import files  
uploaded = files.upload()
```

```
# Get uploaded file name  
file_name = list(uploaded.keys())[0]
```

```
# --- 2. Import Libraries ---  
import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt
```

```
from sklearn.ensemble import RandomForestClassifier  
from sklearn.tree import DecisionTreeClassifier  
from sklearn.linear_model import LogisticRegression  
from sklearn.preprocessing import MinMaxScaler  
from sklearn.model_selection import train_test_split  
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
```

```
import warnings
warnings.filterwarnings('ignore')

# --- 3. Load Dataset ---
if file_name.endswith('.csv'):
    df = pd.read_csv(file_name)
else:
    df = pd.read_excel(file_name)

print("Dataset Loaded:\n", df.head())

# --- 4. Feature Selection ---
features = ['pH', 'EC', 'TDS', 'Alkalinity', 'Hardness', 'Cl', 'F', 'NO3', 'SO4']

X = df[features]
y = df['Water_Quality']

# --- 5. Preprocessing ---
scaler = MinMaxScaler()
X_scaled = scaler.fit_transform(X)

# --- 6. Train-Test Split ---
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

```
# --- 7. Model Training ---
models = {
    "Random Forest": RandomForestClassifier(),
    "Decision Tree": DecisionTreeClassifier(),
    "Logistic Regression": LogisticRegression()
}

results = {}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
    results[name] = acc

    print(f"\n{name} Results:")
    print("Accuracy:", acc)
    print("Confusion Matrix:\n",
          confusion_matrix(y_test, y_pred))
    print("Classification Report:\n",
          classification_report(y_test, y_pred))

# --- 8. Best Model ---
best_model = max(results, key=results.get)
print("\nBest Model:", best_model)
```

```
# --- 9. Visualization ---
```

```
# 1. Water Quality Distribution
```

```
plt.figure()
df['Water_Quality'].value_counts().plot(kind='bar')
plt.title("Water Quality Distribution")
plt.show()
```

```
# 2. Correlation Heatmap
```

```
plt.figure()
corr = df.corr()
plt.imshow(corr)
plt.colorbar()
plt.xticks(range(len(corr.columns)), corr.columns, rotation=90)
plt.yticks(range(len(corr.columns)), corr.columns)
plt.title("Correlation Heatmap")
plt.show()
```

```
# 3. Feature Importance
```

```
best = models[best_model]
importances = best.feature_importances_

plt.figure()
plt.bar(features, importances)
plt.xticks(rotation=45)
plt.title("Feature Importance (" + best_model + ")")
plt.show()
```

```
# --- 10. User Input for Prediction ---
print("\nEnter Water Quality Parameters:")
```

```
ph = float(input("pH: "))
ec = float(input("EC: "))
tds = float(input("TDS: "))
alk = float(input("Alkalinity: "))
hard = float(input("Hardness: "))
cl = float(input("Cl: "))
f = float(input("F: "))
no3 = float(input("NO3: "))
so4 = float(input("SO4: "))
```

```
sample = [[ph, ec, tds, alk, hard, cl, f, no3, so4]]
sample_scaled = scaler.transform(sample)
```

```
prediction = models[best_model].predict(sample_scaled)
```

```
print("\nPrediction:",
      "Safe Water" if prediction[0] == 1 else "Unsafe Water")
```

OUTPUT

Random Forest Results:

Accuracy: 0.95

Confusion Matrix:

[[57 0]

[3 0]]

Classification Report:

	precision	recall	f1-score	support
0	0.95	1.00	0.97	57
1	0.00	0.00	0.00	3
accuracy			0.95	60
macro avg	0.47	0.50	0.49	60
weighted avg	0.90	0.95	0.93	60

Decision Tree Results:

Accuracy: 0.9666666666666667

Confusion Matrix:

[[57 0]

[2 1]]

Classification Report:

	precision	recall	f1-score	support
0	0.97	1.00	0.98	57
1	1.00	0.33	0.50	3
accuracy			0.97	60
macro avg	0.98	0.67	0.74	60
weighted avg	0.97	0.97	0.96	60

Logistic Regression Results:

Accuracy: 0.95

Confusion Matrix:

[[57 0]

[3 0]]

Classification Report:

	precision	recall	f1-score	support
0	0.95	1.00	0.97	57
1	0.00	0.00	0.00	3
accuracy			0.95	60
macro avg	0.47	0.50	0.49	60
weighted avg	0.90	0.95	0.93	60

Best Model: Decision Tree

DISCUSSION

Key Insights Derived from the Project

1. accuracy in classifying groundwater quality based on physicochemical parameters.
- 2.Parameters like TDS, EC, and hardness had strong influence on water quality prediction.

Challenges Faced (Data Quality, Noise, Imbalance, Feature Selection)

- 1.Presence of missing values and noisy data affected model performance.
- 2.Class imbalance (more “safe” than “unsafe” samples) required careful handling for accurate prediction.

Limitations of the Approach or Dataset

- 1.Limited dataset size and region-specific data reduce generalization to other areas.
- 2.Lack of temporal data (seasonal variations) limits long-term prediction capability.

Impact on Agricultural Decision-Making

- 1.Helps farmers identify safe groundwater for irrigation, improving crop yield and soil health.
- 2.Supports efficient water resource management and sustainable farming

CONCLUSION & FUTUREWORK

Project Summary & Outcomes

- Random Forest performed best, providing reliable and accurate predictions.

Potential Improvements

- Use larger and more diverse datasets (multi-seasonal, multi-location).
- Apply advanced models (Deep Learning, Ensemble techniques) and real-time IoT-based monitoring systems.

Practical Applications in Farming

- Helps farmers select suitable water for irrigation, preventing soil degradation.
- Supports precision agriculture and sustainable water management.

Learning & Acknowledgement

- Gained hands-on experience in data preprocessing, model building, and evaluation.
- Developed skills in applying AI/ML for real-world agricultural problems.

REFERENCES

DATA SOURCE : Ground water quality
2018 post monsoon data from kaggle

1. Devendra Dohare, Shriram Deshpande, Atul Kotiya -Research Journal of Engineering Sciences ISSN 2278, 9472, 2014
2. Praharsh S Patel, Dishant M Pandya, Manan Shah(2023)- Environmental Science and Pollution Research 30 (19), 54303-54323, 2023
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THANK YOU !

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